

Analyzing and Recording Transactions

Chapter 2

Chapter 2 Learning Objectives

CONCEPTUAL

- C1** Explain the steps in processing transactions and the role of source documents.
- C2** Describe an account and its use in recording transactions.
- C3** Describe a ledger and a chart of accounts.
- C4** Define *debts* and *credits* and explain **double-entry accounting**.

ANALYTICAL

- A1** Analyze the impact of transactions on accounts and financial statements.
- A2** Compute the debt ratio and describe its use in analyzing financial condition.

PROCEDURAL

- P1** Record transactions in a journal and post entries to a ledger.
- P2** Prepare and explain the use of a trial balance.
- P3** Prepare financial statements from business transactions.

Learning Objective

C1:

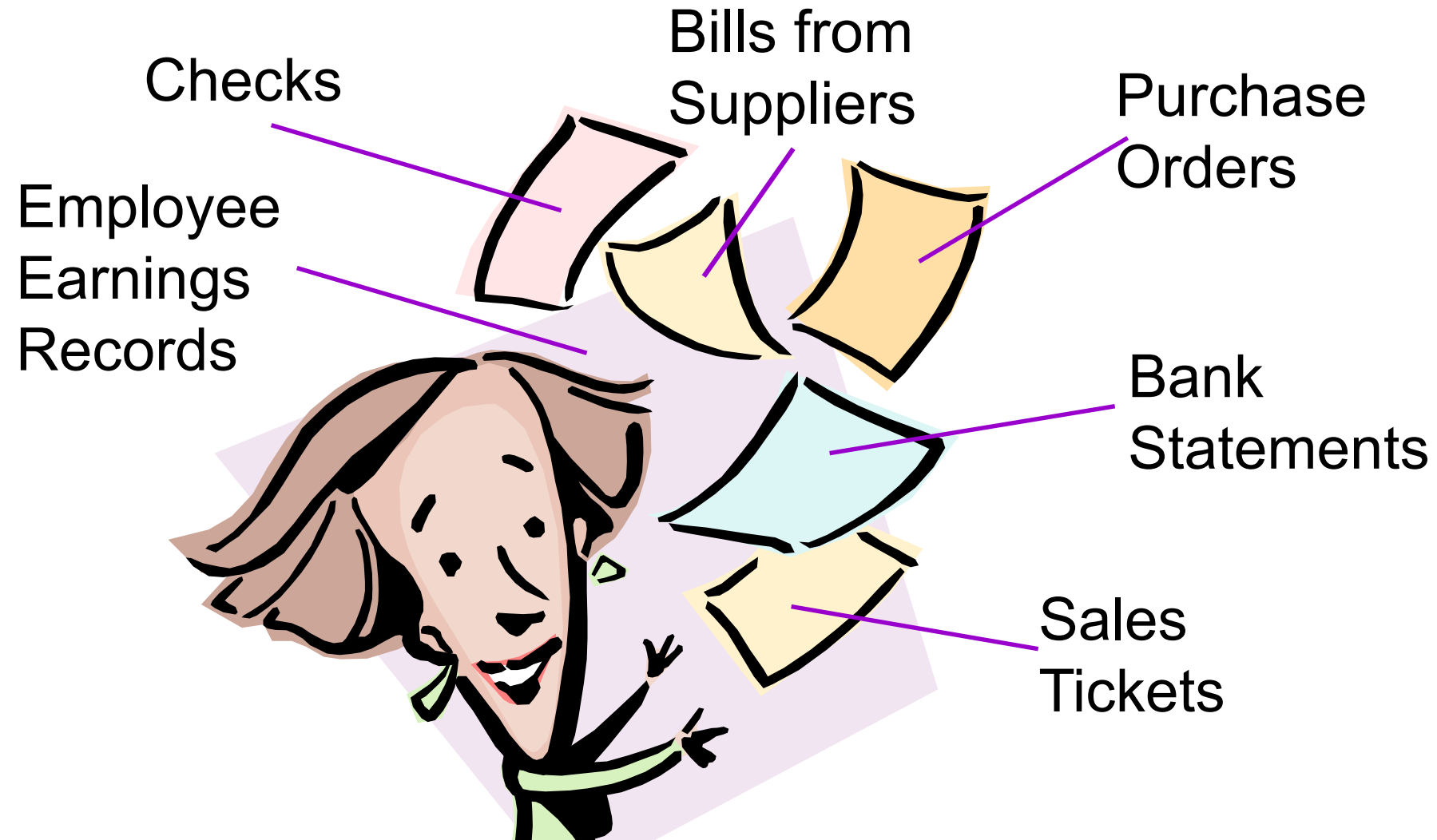
Explain the steps in processing transactions and the role of source documents.

System of Accounts

Business transactions and events are the starting points of financial statements. Process from transactions to financial statements is as follows:

- Identify each transaction and event from source documents.
- Analyze each transaction and event using the accounting equation.
- Record relevant transactions and events in a journal.
- Post journal information to ledger accounts.
- Prepare and analyze the trial balance and financial statements.

Source Documents



Learning Objective

C2:

Describe an account and its use in recording transactions.

The Account and Its Analysis

An **account** is a record of increases and decreases in a specific asset, liability, equity, revenue, or expense.

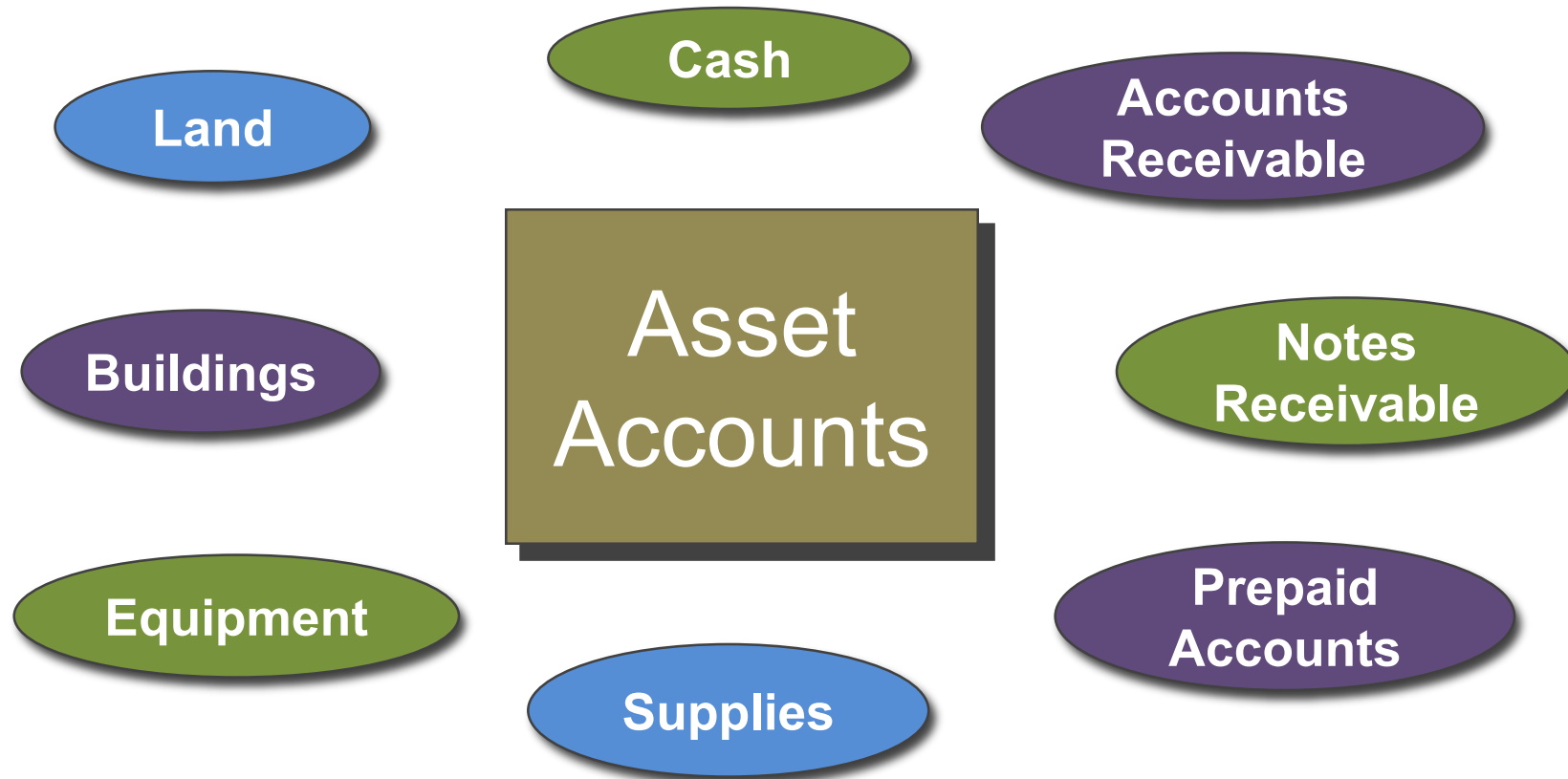
The **general ledger** is a record of all accounts used by the company.

The Account and Its Analysis

Exhibit
2.1



Asset Accounts



Liability Accounts

**Accounts
Payable**

**Notes
Payable**

**Liability
Accounts**

**Accrued
Liabilities**

**Unearned
Revenue**

Equity Accounts

+
**Owner's
Capital**

-
**Owner's
Withdrawals**

**Equity
Accounts**

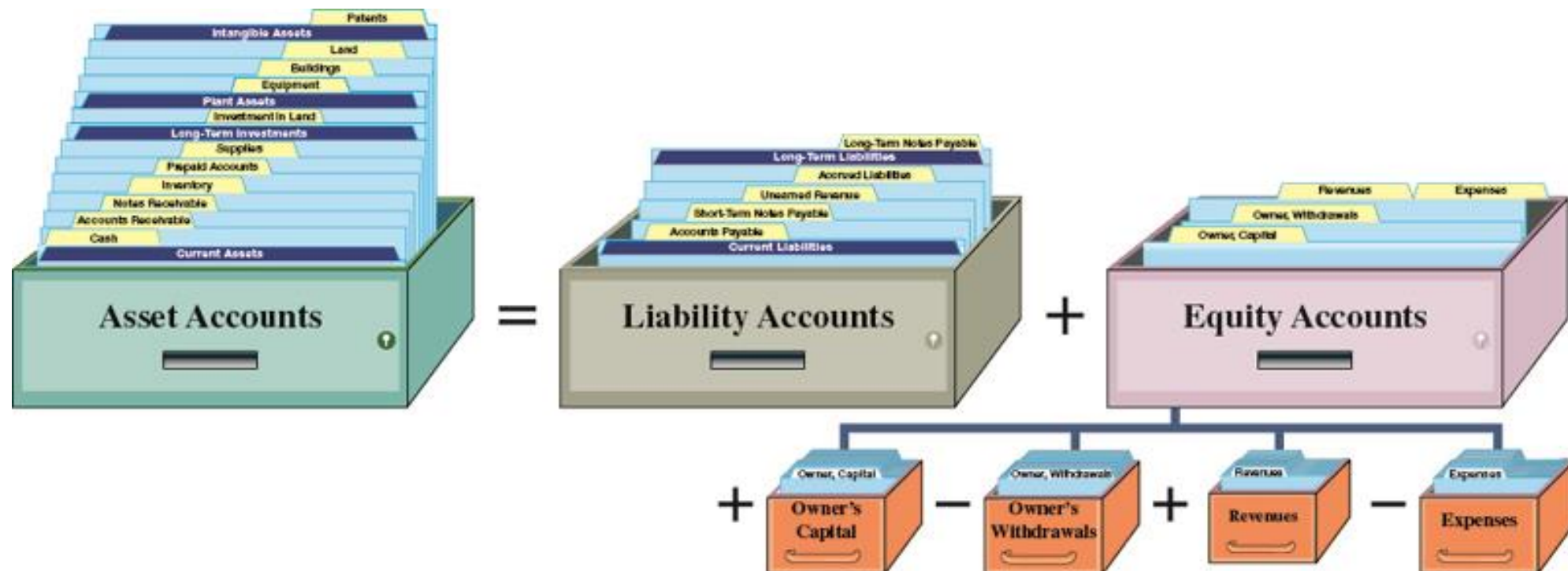
+
Revenues

-
Expenses

The Account and Its Analysis

Revenues and owner's contributions increase equity.
Expenses and owner's withdrawals decrease equity.

Exhibit
2.2



Learning Objective

C3:

Describe a ledger and chart of accounts.

Ledger and Chart of Accounts

The ledger is a collection of all accounts for an accounting system. A company's size and diversity of operations affect the number of accounts needed.

The chart of accounts is a list of all accounts and includes an identifying number for each account.

Exhibit 2.4

Chart of Accounts					
Assets		Liabilities		Equity	
101	Cash	201	Accounts payable	301	C. Taylor, Capital
106	Accounts receivable	236	Unearned consulting revenue	302	C. Taylor, Withdrawals
126	Supplies			Revenues	Expenses
128	Prepaid insurance			403 Consulting revenue	622 Salaries expense
167	Equipment			406 Rental revenue	637 Insurance expense
					640 Rent expense
					652 Supplies expense
					690 Utilities expense

Learning Objective

C4:

Define debits and credits and
explain double-entry
accounting.

Debits and Credits

A T-account represents a ledger account and is used to depict the effects of one or more transactions.

Account Title	
(Left side) Debit	(Right side) Credit

Exhibit
2.5

Double-Entry Accounting

$$\boxed{\text{Assets}} = \boxed{\text{Liabilities}} + \boxed{\text{Equity}}$$

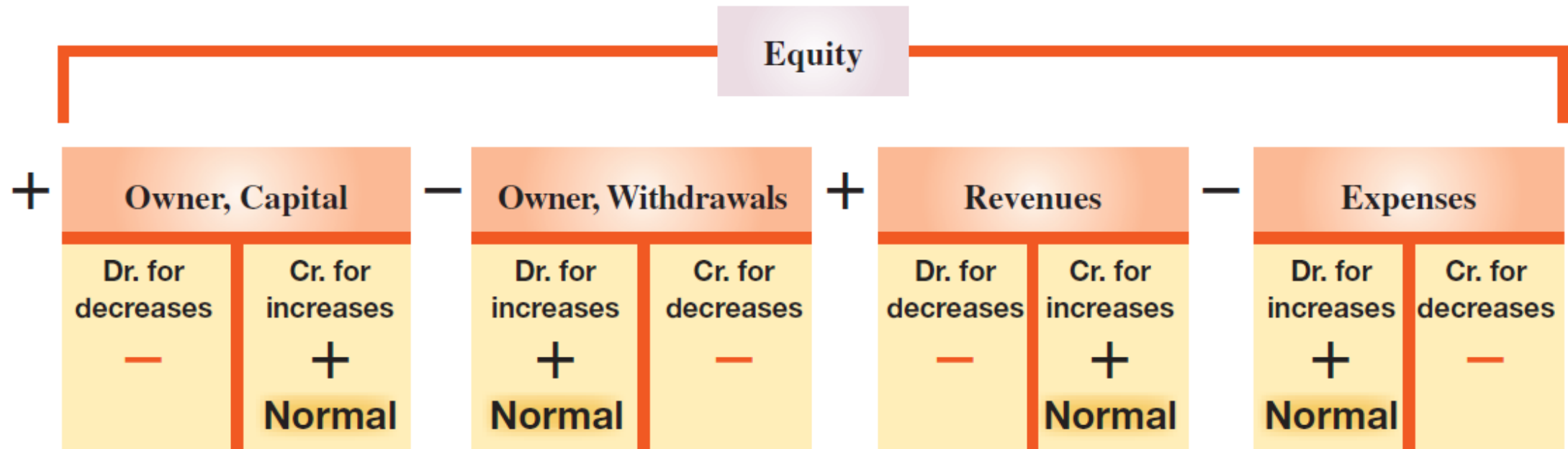
Exhibit
2.6

Assets		=	Liabilities		+	Equity	
Debit for increases +	Credit for decreases -		Debit for decreases -	Credit for increases +		Debit for decreases -	Credit for increases +
Normal				Normal			Normal

Double-Entry Accounting

Here is the expanded accounting equation showing the equity section.

Exhibit
2.7



Double-Entry Accounting

An account balance is the difference between the increases and decreases in an account. Notice the T-Account.

Exhibit
2.8

Cash			
36,100	{	Receive investment by owner	30,000
		Consulting services revenue earned	4,200
		Collection of account receivable	1,900
		Purchase of supplies	2,500
		Purchase of equipment	26,000
		Payment of rent	1,000
		Payment of salary	700
		Payment of account payable	900
		Withdrawal by owner	200
		31,300	
Balance		4,800	
36,100 – 31,300			

Learning Objective

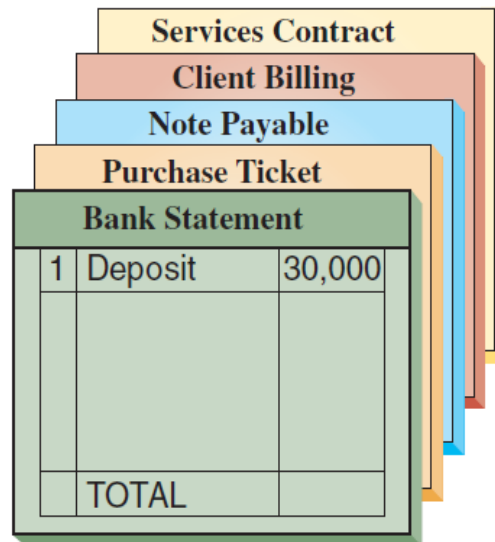
P1:

Record transactions in a journal and post entries to a ledger.

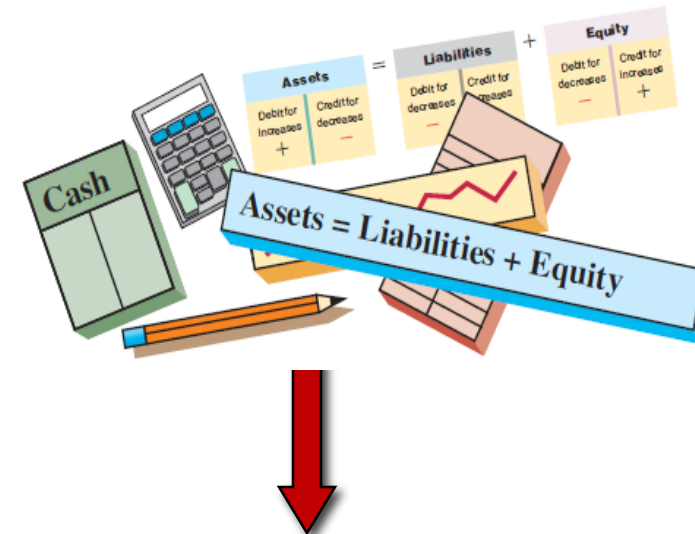
Journalizing and Posting Transactions

Exhibit
2.9

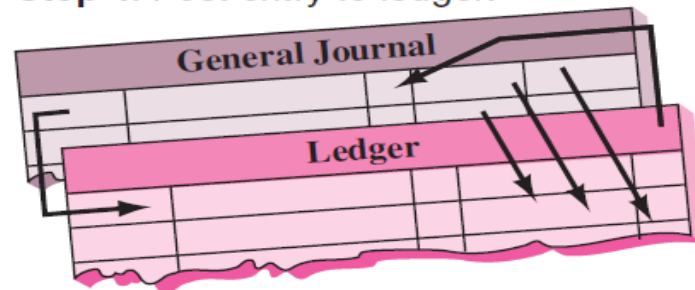
Step 1: Identify transactions and source documents.



Step 2: Analyze transactions using the accounting equation.



Step 4: Post entry to ledger.



Step 3: Record journal entry.

General Journal			
Dec. 1	Cash	30,000	
	Taylor, Capital		30,000
Dec. 2	Supplies	2,500	
	Cash		2,500

Journalizing Transactions

Exhibit
2.10

a. Transaction
Date

b. Titles of Affected
Accounts

General Journal				
Date: Dec 1, 2017 Reference: Original Transaction: ☐				
Date	Account Titles and Explanation	PR	Debit	Credit
2017 (a) Dec. 1	(b) Cash		30,000	
	C. Taylor, Capital			(c) 30,000
	Receive investment by owner. (d)			
Dec. 2	Supplies		2,500	
	Cash			2,500
	Purchase supplies for cash.			

d. Transaction
explanation

c. Dollar amount of debits
and credits

Balance Account Column

Exhibit
2.11

T-accounts are useful illustrations, but **balance column ledger accounts** are used in practice.

General Ledger					
Cash				Account No. 101	
Date	Explanation	PR	Debit	Credit	Balance
2017 Dec. 1		G1	30,000		30,000
Dec. 2		G1		2,500	27,500
Dec. 3		G1		26,000	1,500
Dec. 10		G1	4,200		5,700

Posting Journal Entries

Exhibit
2.12

Peachtree Accounting: FastForward

File Edit Lists Maintain Tasks Analysis Options Reports & Forms Services Window Help

Company

- Business Status
- Customers & Sales
- Vendors & Purchases
- Inventory & Services
- Employees & Payroll
- Banking
- Company

Shortcuts [Customize](#)

- Sales Invoice
- Receive Money from Customer
- Bills - Pay Bill
- Customer List
- Vendor List
- Find Transactions
- General Journal Entry

General Journal Entry

File Edit Go To Window Help

Close New Lpt Save Print Delete Recog Row Reports Help

General Journal

Date: Dec 1, 2015 Reference: Reverse Transaction

Date	Account Titles and Explanation	PR	Debit	Credit
2015 Dec. 1	Cash	101	30,000	
	C. Taylor, Capital	301		30,000
	Receive investment by owner.			

General Ledger

File Edit Go To Window Help

Close Print Print Options Columns Fields Setup Hide Email Export PDF Print Help

Cash Account no. 101

Date	Explanation	PR	Debit	Credit	Balance
2015 Dec. 1		G1	30,000		30,000

C. Taylor, Capital Account no. 301

Date	Explanation	PR	Debit	Credit	Balance
2015 Dec. 1		G1		30,000	30,000

Diagram illustrating the posting of journal entries to the ledger:

1. Select the General Journal Entry from the Shortcuts menu.
2. Select the entry to be posted.
3. Select the account number (101) for Cash.
4. Select the account number (301) for C. Taylor, Capital.

Learning Objective P1: Record transactions in a journal and post entries to a ledger.

Learning Objective

A1:

Analyze the impact of transactions on accounts and financial statements.

Processing Transactions

Double-entry accounting is useful in analyzing and processing transactions. Analysis of each transaction follows these four steps.

- Step 1** Identify the transaction and any source documents.
- Step 2** Analyze the transaction using the accounting equation.
- Step 3** Record the transaction in journal entry form applying double-entry accounting.
- Step 4** Post the entry (for simplicity, we use T-accounts to represent ledger accounts).

Processing Transactions

1. Receive investment by Owner

1 IDENTIFY FastForward receives \$30,000 cash from Chas Taylor as an owner contribution.

2 ANALYZE

Assets	=	Liabilities	+	Equity
Cash				C. Taylor, Capital
+30,000	=	0		+30,000

3 RECORD

Date	Account Titles and Explanation	PR	Debit	Credit
(1)	Cash	101	30,000	
	C. Taylor, Capital	301		30,000

4 POST

Cash		101
(1)	30,000	

C. Taylor, Capital		301
	(1)	30,000

Processing Transactions

2. Purchase Supplies for Cash

1 IDENTIFY FastForward pays \$2,500 cash for supplies.

2 ANALYZE

Assets		=	Liabilities	+	Equity
Cash	Supplies				
-2,500	+2,500	=	0	+	0

Changes the composition of assets but not the total.

3 RECORD

Date	Account Titles and Explanation	PR	Debit	Credit
(2)	Supplies	126	2,500	
	Cash	101		2,500

4 POST

Supplies		126
(2)	2,500	

Cash		101
(1)	30,000	(2) 2,500

Processing Transactions

3. Purchase Equipment for Cash

1 IDENTIFY FastForward pays \$26,000 cash for equipment.

2 ANALYZE

Assets		=	Liabilities	+	Equity
Cash	Equipment				
-26,000	+26,000	=	0	+	0

Changes the composition of assets but not the total.

3 RECORD

Date	Account Titles and Explanation	PR	Debit	Credit
(3)	Equipment	167	26,000	
	Cash	101		26,000

4 POST

Equipment		167
(3)	26,000	

Cash		101
(1)	30,000	(2) 2,500
		(3) 26,000

Processing Transactions

4. Purchase Supplies on Credit

1 IDENTIFY FastForward purchases \$7,100 of supplies on credit from a supplier.

2 ANALYZE

Assets	=	Liabilities	+	Equity
Supplies		Accounts Payable		
+7,100	=	+7,100	+	0

3 RECORD

Date	Account Titles and Explanation	PR	Debit	Credit
(4)	Supplies	126	7,100	
	Accounts Payable	201		7,100

4 POST

Supplies		126
(2)	2,500	
(4)	7,100	

Accounts Payable		201
	(4)	7,100

Processing Transactions

5. Provide Services for Cash

1 IDENTIFY FastForward provides consulting services and immediately collects \$4,200 cash.

2 ANALYZE

Assets	=	Liabilities	+	Equity
Cash				Consulting Revenue
+4,200	=	0		+4,200

3 RECORD

Date	Account Titles and Explanation	PR	Debit	Credit
(5)	Cash	101	4,200	
	Consulting Revenue	403		4,200

4 POST

Cash		101
(1)	30,000	(2) 2,500
(5)	4,200	(3) 26,000

Consulting Revenue		403
	(5) 4,200	

Processing Transactions

6. Payment of Expense in Cash

1 IDENTIFY FastForward pays \$1,000 cash for December rent.

2 ANALYZE

Assets	=	Liabilities	+	Equity
Cash				Rent Expense
- 1,000	=	0		- 1,000

3 RECORD

Date	Account Titles and Explanation	PR	Debit	Credit
(6)	Rent Expense	640	1,000	
	Cash	101		1,000

4 POST

	Rent Expense	640
(6)	1,000	

	Cash	101
(1)	30,000	(2) 2,500
(5)	4,200	(3) 26,000
		(6) 1,000

Processing Transactions

7. Payment of Expense in Cash

1 IDENTIFY FastForward pays \$700 cash for employee salary.

2 ANALYZE

Assets	=	Liabilities	+	Equity
Cash				Salaries Expense
-700	=	0		-700

3 RECORD

Date	Account Titles and Explanation	PR	Debit	Credit
(7)	Salaries Expense	622	700	
	Cash	101		700

4 POST

Salaries Expense		622
(7)	700	

Cash		101
(1)	30,000	(2) 2,500
(5)	4,200	(3) 26,000
		(6) 1,000
		(7) 700

Processing Transactions

8. Provide Consulting and Rental Services on Credit

1 IDENTIFY FastForward provides consulting services of \$1,600 and rents its test facilities for \$300. The customer is billed \$1,900 for these services.

2 ANALYZE

Assets	=	Liabilities	+	Equity
Accounts Receivable				Consulting Revenue Rental Revenue
+1,900	=	0		+1,600 +300

3 RECORD

Date	Account Titles and Explanation	PR	Debit	Credit
(8)	Accounts Receivable	106	1,900	
	Consulting Revenue	403		1,600
	Rental Revenue	406		300

4 POST

Accounts Receivable		106
(8)	1,900	

Consulting Revenue		403
	(5)	4,200
	(8)	1,600

Rental Revenue		406
	(8)	300

Processing Transactions

9. Receipt of Cash on Account

1 IDENTIFY FastForward receives \$1,900 cash from the client billed in transaction 8.

2 ANALYZE

Assets		=	Liabilities	+	Equity
Cash	Accounts Receivable				
+1,900	-1,900	=	0	+	0

3 RECORD

Date	Account Titles and Explanation	PR	Debit	Credit
(9)	Cash	101	1,900	
	Accounts Receivable	106		1,900

4 POST

Cash		101	
(1)	30,000	(2)	2,500
(5)	4,200	(3)	26,000
→ (9)	1,900	(6)	1,000
		(7)	700

Accounts Receivable		106
(8)	1,900	(9) 1,900

Processing Transactions

10. Partial Payment of Accounts Payable

1 IDENTIFY FastForward pays CalTech Supply \$900 cash toward the payable of transaction 4.

2 ANALYZE

Assets	=	Liabilities	+	Equity
Cash		Accounts Payable		
-900	=	-900	+	0

3 RECORD

Date	Account Titles and Explanation	PR	Debit	Credit
(10)	Accounts Payable	201	900	
	Cash	101		900

4 POST

Accounts Payable 201	
(10) 900	(4) 7,100

Cash 101	
(1) 30,000	(2) 2,500
(5) 4,200	(3) 26,000
(9) 1,900	(6) 1,000
	(7) 700
	(10) 900

Processing Transactions

11. Withdrawal of Cash by Owner

1 IDENTIFY Chas Taylor withdraws \$200 cash from FastForward for personal use.

2 ANALYZE

Assets	=	Liabilities	+	Equity
Cash				C. Taylor, Withdrawals
-200	=	0		-200

3 RECORD

Date	Account Titles and Explanation	PR	Debit	Credit
(11)	C. Taylor, Withdrawals	302	200	
	Cash	101		200

4 POST

C. Taylor, Withdrawals		302
(11)	200	

Cash		101
(1)	30,000	(2) 2,500
(5)	4,200	(3) 26,000
(9)	1,900	(6) 1,000
		(7) 700
		(10) 900
		(11) 200

Processing Transactions

12. Receipt of Cash for Future Services

1 IDENTIFY FastForward receives \$3,000 cash in advance of providing consulting services to a customer.

2 ANALYZE

Assets	=	Liabilities	+	Equity
Unearned Consulting Revenue				
Cash	=	Unearned Consulting Revenue	+	
+3,000	=	+3,000	+	0

Accepting \$3,000 cash obligates FastForward to perform future services and is a liability. No revenue is earned until services are provided.

3 RECORD

Date	Account Titles and Explanation	PR	Debit	Credit
(12)	Cash	101	3,000	
	Unearned Consulting Revenue	236		3,000

4 POST

Cash		101
(1)	30,000	(2) 2,500
(5)	4,200	(3) 26,000
(9)	1,900	(6) 1,000
(12)	3,000	(7) 700
		(10) 900
		(11) 200

Unearned Consulting Revenue		236
	(12)	3,000

Processing Transactions

13. Pay Cash for Future Insurance Coverage

1 IDENTIFY FastForward pays \$2,400 cash (insurance premium) for a 24-month insurance policy. Coverage begins on December 1.

2 ANALYZE

Assets		=	Liabilities	+	Equity
Cash	Prepaid Insurance				
-2,400	+2,400	=	0	+	0

Changes the composition of assets from cash to prepaid insurance. Expense is incurred as insurance coverage expires.

3 RECORD

Date	Account Titles and Explanation	PR	Debit	Credit
(13)	Prepaid Insurance	128	2,400	
	Cash	101		2,400

4 POST

Prepaid Insurance		128
(13)	2,400	

Cash		101
(1)	30,000	(2) 2,500
(5)	4,200	(3) 26,000
(9)	1,900	(6) 1,000
(12)	3,000	(7) 700
		(10) 900
		(11) 200
		(13) 2,400

Processing Transactions

14. Purchase Supplies for Cash

1 IDENTIFY FastForward pays \$120 cash for supplies.

2 ANALYZE

Assets		=	Liabilities	+	Equity
Cash	Supplies				
-120	+120	=	0	+	0

3 RECORD

Date	Account Titles and Explanation	PR	Debit	Credit
(14)	Supplies	126	120	
	Cash	101		120

4 POST

Supplies		126
(2)	2,500	
(4)	7,100	
(14)	120	

Cash		101
(1)	30,000	(2) 2,500
(5)	4,200	(3) 26,000
(9)	1,900	(6) 1,000
(12)	3,000	(7) 700
		(10) 900
		(11) 200
		(13) 2,400
		(14) 120

Processing Transactions

15. Payment of Expense in Cash

1 IDENTIFY FastForward pays \$305 cash for December utilities expense.

2 ANALYZE

Assets	=	Liabilities	+	Equity
Cash				Utilities Expense
-305	=	0		-305

3 RECORD

Date	Account Titles and Explanation	PR	Debit	Credit
(15)	Utilities Expense	690	305	
	Cash	101		305

4 POST

Utilities Expense		690
(15)	305	

Cash		101	
(1)	30,000	(2)	2,500
(5)	4,200	(3)	26,000
(9)	1,900	(6)	1,000
(12)	3,000	(7)	700
		(10)	900
		(11)	200
		(13)	2,400
		(14)	120
		(15)	305

Processing Transactions

16. Payment of Expense in Cash

1 IDENTIFY FastForward pays \$700 cash in employee salary for work performed in the latter part of December.

2 ANALYZE

Assets	=	Liabilities	+	Equity
Cash				Salaries Expense
-700	=	0		-700

3 RECORD

Date	Account Titles and Explanation	PR	Debit	Credit
(16)	Salaries Expense	622	700	
	Cash	101		700

4 POST

Salaries Expense		622
(7)	700	
(16)	700	

Cash		101
(1)	30,000	(2) 2,500
(5)	4,200	(3) 26,000
(9)	1,900	(6) 1,000
(12)	3,000	(7) 700
		(10) 900
		(11) 200
		(13) 2,400
		(14) 120
		(15) 305
		(16) 700

Exhibit 2.13

Learning Objective A1: Analyze the impact of transactions on accounts and financial statements.

Learning Objective

P2:

Prepare and explain the
use of a trial balance.

Preparing a Trial Balance

Preparing a trial balance involves three steps:

1. List each account title and its amount (from ledger) in the trial balance. If an account has a zero balance, list it with a zero in the normal balance column (or omit it entirely).
2. Compute the total of debit balances and the total of credit balances.
3. Verify (prove) total debit balances equal total credit balances.

After processing its remaining transactions for December, FastForward's Trial Balance is prepared.

FAST FORWARD Trial Balance December 31, 2017		
	Debit	Credit
Cash	\$ 4,275	
Accounts receivable	0	
Supplies	9,720	
Prepaid insurance	2,400	
Equipment	26,000	
Accounts payable		\$ 6,200
Unearned consulting revenue		3,000
C. Taylor, Capital		30,000
C. Taylor, Withdrawals	200	
Consulting revenue		5,800
Rental revenue		300
Salaries expense	1,400	
Rent expense	1,000	
Utilities expense	305	
Totals	\$ 45,300	\$ 45,300

The trial balance lists all ledger accounts and their balances at a point in time. If the books are in balance, the total debits will equal the total credits.

Searching for Errors

If the trial balance does not balance, the error(s) must be found and corrected.

① Make sure the trial balance columns are correctly added.

④ Re-compute each account balance in the ledger.

② Make sure account balances are correctly entered from the ledger.

⑤ Verify that each journal entry is posted correctly.

③ See if debit or credit accounts are mistakenly placed on the trial balance.

⑥ Verify that each original journal entry has equal debits and credits.

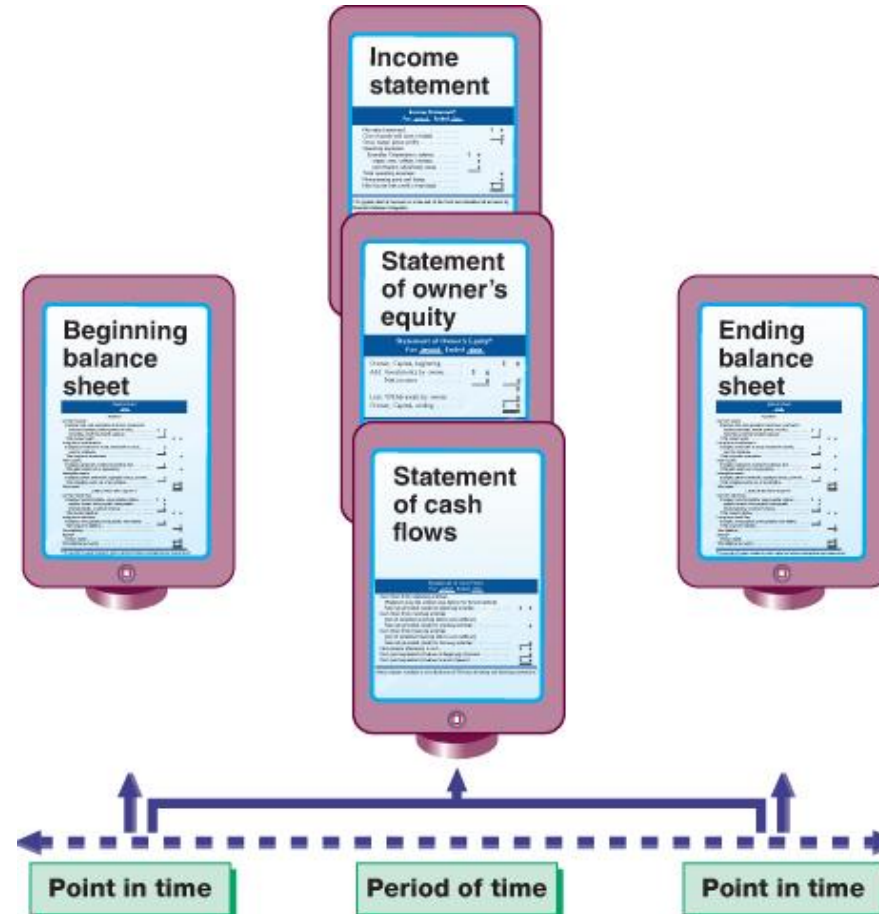
Learning Objective

P3:

Prepare financial statements
from business transactions.

Financial Statements Prepared from Trial Balance

Exhibit
2.15



Financial Statements

The four financial statements and their purposes are:

- 1. Income statement** — reports revenues less expenses along with the resulting net income or loss over a period of time due to earnings activities.
- 2. Statement of owner's equity**— reports how equity changes over the reporting period from net income (or loss) and from any owner investments and withdrawals over a period of time.
- 3. Balance sheet** — reports the financial position (types and amounts of assets, liabilities, and equity) at a point in time.
- 4. Statement of Cash Flows** — The statement of cash flows lists the cash inflows and cash outflows for the period.

For simplicity, we do not show the statement of cash flows for FastForward in this chapter, but we do return to this statement in the next chapter.

Income Statement

Exhibit
2.16

FASTFORWARD Income Statement For Month Ended December 31, 2017		
Revenues		
Consulting revenue (\$4,200 + \$1,600)	\$ 5,800	
Rental revenue	<u>300</u>	
Total revenues		\$ 6,100
Expenses		
Salaries expense	1,400	
Rent expense	1,000	
Utilities expense	<u>305</u>	
Total expenses		<u>2,705</u>
Net income		<u><u>\$ 3,395</u></u>

Statement of Owner's Equity

Exhibit
2.16

FASTFORWARD Income Statement For Month Ended December 31, 2017		
Revenues		
Consulting revenue (\$4,200 + \$1,600)	\$ 5,800	
Rental revenue	300	
Total revenues		\$ 6,100
Expenses		
Salaries expense	1,400	
Rent expense	1,000	
Utilities expense	305	
Total expenses		2,705
Net income		<u><u>\$ 3,395</u></u>

FASTFORWARD Statement of Owner's Equity For Month Ended December 31, 2017		
C. Taylor, Capital, December 1, 2017		\$ 0
→ Plus: Investments by owner	\$30,000	
Net income	<u>3,395</u>	33,395
		33,395
→ Less: Withdrawals by owner		200
C. Taylor, Capital, December 31, 2017		<u><u>\$33,195</u></u>

Balance Sheet

Exhibit
2.16

FASTFORWARD Statement of Owner's Equity For Month Ended December 31, 2017			
C. Taylor, Capital, December 1, 2017		\$	0
➔ Plus: Investments by owner	\$30,000		
Net income	<u>3,395</u>		<u>33,395</u>
			33,395
➔ Less: Withdrawals by owner			<u>200</u>
C. Taylor, Capital, December 31, 2017			<u><u>\$33,195</u></u>

FASTFORWARD Balance Sheet December 31, 2017			
Assets		Liabilities	
Cash	\$ 4,275	Accounts payable	\$ 6,200
Supplies	9,720	Unearned consult. revenue	<u>3,000</u>
Prepaid insurance ...	2,400	Total liabilities	<u>9,200</u>
Equipment	26,000	Equity	
		C. Taylor, Capital	<u>33,195</u>
Total assets	<u><u>\$42,395</u></u>	Total liabilities and equity	<u><u>\$ 42,395</u></u>

Presentation Issues

1. Dollar signs are not used in journals and ledgers.
2. Dollar signs appear in financial statements and other reports such as trial balances. The usual practice is to put dollar signs beside only the first and last numbers in a column.
3. When amounts are entered in the journal, ledger, or trial balance, commas are optional to indicate thousands, millions, and so forth.
4. Commas are always used in financial statements.
5. Companies commonly round amounts in reports to the nearest dollar, or even to a higher level.

Learning Objective

A2:

Compute the debt ratio and describe its use in analyzing financial condition.

Debt Ratio

$$\text{Debt Ratio} = \frac{\text{Total Liabilities}}{\text{Total Assets}}$$

Evaluates the level of debt risk.

A higher ratio indicates that there is a greater probability that a company will not be able to pay its debt in the future.

Debt Ratio

Exhibit
2.18

$$\text{Debt Ratio} = \frac{\text{Total Liabilities}}{\text{Total Assets}}$$

\$ millions	2015	2014	2013	2012	2011
Total liabilities.....	\$ 672	\$ 541	\$ 429	\$ 421	\$ 389
Total assets.....	\$2,047	\$1,675	\$1,409	\$1,340	\$1,282
Debt ratio	0.33	0.32	0.30	0.31	0.30
Industry debt ratio	0.49	0.49	0.47	0.46	0.47

End of Chapter 2